



IDEA



IDEA / SOLUTION:

A **Blockchain-based Product Ownership Management System** for real-time anti-counterfeit detection after the sale. Counterfeits are about 2.5% of world trade (OECD/EUIPO).

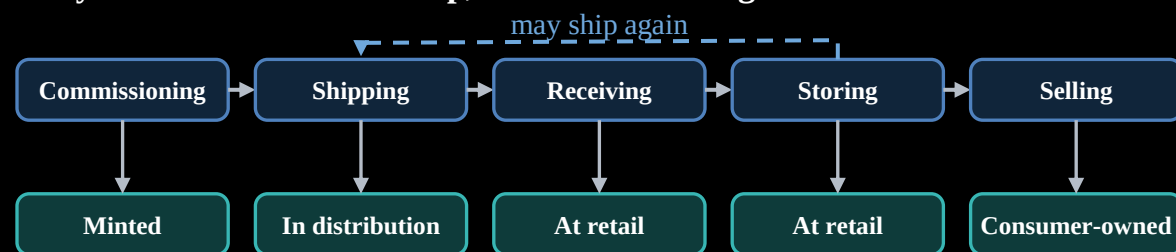
NFT-based Digital Twin: each serialized product is an ERC-721 token. Its ID is derived from GTIN + serial, so a serial can only ever be minted once.

GS1 EPCIS event trail: every handoff is logged with who, what, when and where; the registry anchors only the coarse stage on-chain.

Wallet-signed ownership proof: the owner signs a single-use challenge; no private data is revealed. zk-SNARK privacy proofs are on the roadmap.

Scan to verify: anyone scans the QR code and sees authenticity, provenance and a clone-risk score in seconds.

Lifecycle: off-chain EPCIS step, then on-chain stage



A sold item is final. The chain refuses skipped steps and revoked products.

Problem Resolution:

- **Eliminates duplicate serial number fraud** via cryptographic ownership enforcement
- **Prevents post-supply-chain counterfeit entry** through immutable tracking
- **Keeps sensitive data off-chain:** only coarse lifecycle stages are anchored, event details stay in EPCIS records
- **Removes data silos** with one verification layer shared by manufacturers, distributors, retailers and buyers

Unique Value Propositions (UVP):

Digital twin per item using ERC-721 NFTs on smart contracts

Tamper-proof lifecycle ledger with role-based access

Interoperable via GS1 EPCIS and GTIN check-digit validation

Works today: live app, demo token 0xde70a11ce0000001, no wallet needed

Clone detection: every scan is scored for copy-like patterns, so a photocopied label is flagged

Engineered to ship: open source (MIT), 88 unit + 11 browser tests, CI on every change

Live Deployed Product: <https://authen-tick-web.vercel.app>



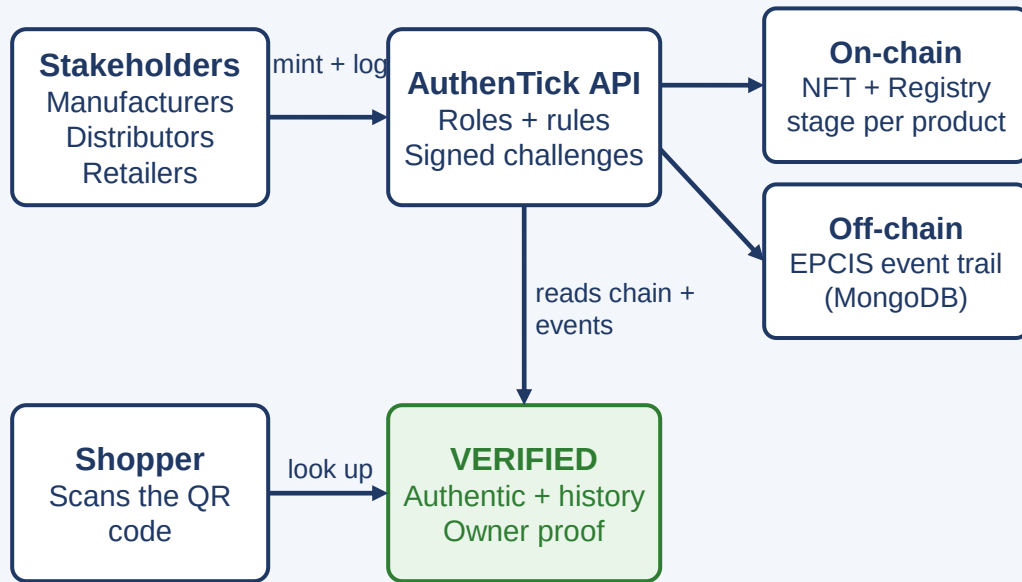
Technical Approach



Github Repository (Documentation and code) : <https://github.com/Phantom8420/AuthenTick>

Ownership & Verification Flow

AuthenTick: blockchain-based product ownership management



Product Status:

Built, deployed and tested: contracts, API and web app are complete and the app is live. A full product life (mint to sale, plus clone and revoke checks) runs against a real local chain in CI. Next: testnet deployment, zk-SNARK proofs, tamper-proof tags.

Algorithm Development:

Ownership and lifecycle logic in smart contracts: an **ERC-721 NFT** per product (ID = keccak256 of GTIN + serial) and an **OwnershipRegistry** that only moves forward one role-checked step at a time.

Application Development: React web app (QR scan, verify, mint, supply-chain timeline) on an Express + MongoDB API for **manufacturers, distributors, retailers and customers**. A built-in demo mode needs no backend.

Encryption and Security: Wallet sign-in with **single-use signed challenges**, httpOnly sessions with CSRF protection, role-based access, rate limiting, signed ownership proofs and scan-anomaly clone detection. zk-SNARKs and PUF tags are planned.

Cloud / System Infrastructure:

Blockchain Platform: Ethereum-compatible (Hardhat, OpenZeppelin, ethers)

Messaging Standard: GS1 EPCIS for supply-chain events

Deployment: Docker and Vercel; GitHub Actions CI (unit, end-to-end, accessibility, Slither)



Security Model



Threats and defences:

- **Cloned serial number:** token ID is derived from GTIN + serial; a second mint is rejected by the contract
- **Photocopied QR label:** scans are scored for copy-like patterns and flagged as a likely clone
- **Replayed signature:** challenges are single use and expire in 5 minutes
- **Stolen session, CSRF:** httpOnly cookie plus an anti-CSRF header on every write
- **Role abuse:** roles enforced in the API and the contracts; stages move one step at a time
- **Recalled product:** revoking it freezes it on-chain, so no further stage can be recorded
- **Data exposure:** only coarse stages on-chain; scan sources stored as hashed networks, never IPs

Measured, not claimed:

Gas per on-chain action (a CI test fails if it rises)

Mint (ERC-721)	196k
Stage 1: Minted	64k
Stage 2: Distribution	47k
Stage 3: At retail	47k
Stage 4: Consumer	47k
Revoke	40k

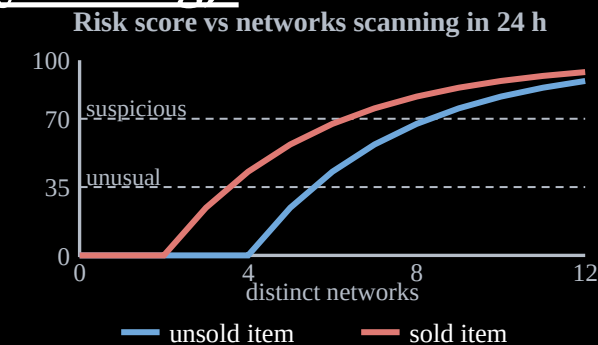
Whole on-chain life (mint + 4 stages): about 400k gas

- **Tests:** 88 unit and 11 browser tests (demo journey, clone flag, accessibility)
- **Checks:** Slither, a real-chain smoke test, CI on every change

Clone detection (anomaly scoring):

A genuine label is read in a few places; a copy shows up everywhere. We log each lookup as a hashed network, score the distinct networks per 24 hours (stricter once sold) and scan bursts, and flag high scores as a likely clone.

Not a black box. Try Simulate a cloned label on the demo item.



Live product : authen-tick-web.vercel.app





Feasibility and Impact



Feasibility Analysis

Technical: Built and tested end to end. Measured: about 400k gas for a product's whole on-chain life.

Financial: Low running cost. Brands pay per mint or subscribe.

Market: Pharma, luxury, electronics. Unlike MediLedger or VeChain: open source, buyer-side checks, live demo.

Operational: Open standards (GS1 EPCIS, ERC-721) fit existing systems.

Challenges & Risks

Technical: Gas fees; the relayer is a trust point; contracts not yet independently audited.

Financial: Implementation and onboarding cost for manufacturers.

Market: Slow industry adoption, resistance to change.

Operational: Linking the item to its token relies on label integrity; data silos.

Mitigation Strategies

Technical: Layer-2 deployment, batch anchoring, independent audit, zk-SNARK proofs.

Financial: Consortium cost-sharing; pay-per-mint model.

Market: Start with one sector pilot; industry partnerships.

Operational: Standardize on GS1 EPCIS, phased rollout, tamper-evident NFC/PUF tags.

Target Impact

Trust: Verified product authenticity

Safety: Eliminates counterfeit risks

Economic: Reduces fraud losses

Access: Transparent resale markets

Solution Benefits

Social

Consumer trust

Safer products

Transparency

Economic

Fraud reduction

Cost efficiency

Market expansion

Environmental

Reduced waste

Efficient tracking

Optimized logistics



Research and References



Primary Sources

Elsevier (Blockchain & supply chain studies)
ScienceDirect (DLT, anti-counterfeit research)

Industry Reports & Pilots

MediLedger Project (Pharma blockchain pilot)
DSCSA Compliance Framework (U.S. drug supply chain)

Technical Standards

GS1 EPCIS (Supply chain data standard)
ERC-721 (NFT standard)
zk-SNARKs (Privacy protocol, planned)

Built With

OpenZeppelin Contracts, Hardhat, ethers.js, React, Express, MongoDB
Source and docs: github.com/Phantom8420/AuthenTick